6406533459031. ✖ All of these

MLF

Section Id :	64065373889
Section Number :	5
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	12
Number of Questions to be attempted :	12
Section Marks :	40
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653155064
Question Shuffling Allowed :	No

Question Number : 64 Question Id : 6406531021289 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING FOUNDATIONS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406533459032. ✔ YES

6406533459033. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653155065
Question Shuffling Allowed :	Yes

Question Number : 65 Question Id : 6406531021290 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider the matrix

$$B = \begin{bmatrix} 4 & -2i & -1 \\ 2i & 0 & -i \\ -1 & i & 4 \end{bmatrix}$$

Which of the following options is/are true with respect to B?

Note: B^H is a conjugate transpose of B.

Options :

6406533459034. ✖ $B^2 = -B^H$

6406533459035. ✔ $B = B^H$

6406533459036. ✔ Every eigenvalue of B is real.

6406533459037. ✖ The determinant of B is zero.

Question Number : 66 Question Id : 6406531021298 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider a unitary matrix A. Let α be an eigenvalue of A, then which of the following options is/are true?

Options :

6406533459062. ✔ $|\alpha| = 1$.

6406533459063. ✔ α is an eigenvalue of A^H, where A^H is a hermitian matrix.

6406533459064. ✔ Eigenvectors of A are orthonormal.

6406533459065. ✖ A is not diagonalizable.

Sub-Section Number :	3
Sub-Section Id :	640653155066
Question Shuffling Allowed :	Yes

Question Number : 67 Question Id : 6406531021291 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Consider the following matrix:

$$A = \begin{bmatrix} 2 & -i & i \\ i & 2 & -i \\ -i & i & 2 \end{bmatrix}$$

Diagonalize it as $A = Q\Lambda Q^T$, then Λ is given by

Options :

6406533459038. ✖ $\begin{bmatrix} 3 & 0 & 0 \\ 0 & 3+\sqrt{2} & 0 \\ 0 & 0 & 3-\sqrt{2} \end{bmatrix}$

6406533459039. ✔ $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2+\sqrt{3} & 0 \\ 0 & 0 & 2-\sqrt{3} \end{bmatrix}$

6406533459040. ✖ $\begin{bmatrix} 3 & 0 & 0 \\ 0 & 2+\sqrt{3} & 0 \\ 0 & 0 & 2-\sqrt{3} \end{bmatrix}$

6406533459041. ✖ The given matrix is not diagonalizable.

Question Number : 68 Question Id : 6406531021296 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Suppose $A = \begin{bmatrix} 1+i & 2 \\ 0 & 3+i \end{bmatrix}$. Then, which of the following options represent the eigenvectors of A?

Options :

6406533459054. ✔ $\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

6406533459055. ✖

$\begin{bmatrix} 0 \\ -i \end{bmatrix}, \begin{bmatrix} i \\ i \end{bmatrix}$

6406533459056. ✖ $\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \end{bmatrix}$

6406533459057. ✖ $\begin{bmatrix} -1 \\ -1 \end{bmatrix}, \begin{bmatrix} -i \\ -i \end{bmatrix}$

Question Number : 69 Question Id : 6406531021297 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Given the quadratic form $Q(x, y, z) = 3x^2 + 2y^2 + z^2 + 4xy + 2xz + 4yz$, expressed in matrix form as $Q(x, y, z) = \begin{bmatrix} x & y & z \end{bmatrix} M \begin{bmatrix} x \\ y \\ z \end{bmatrix}$. Which of the following is M?

Options :

6406533459058. ✔ $\begin{bmatrix} 3 & 2 & 1 \\ 2 & 2 & 2 \\ 1 & 2 & 1 \end{bmatrix}$

6406533459059. ✖ $\begin{bmatrix} 3 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 1 \end{bmatrix}$

6406533459060. ✖ $\begin{bmatrix} 2 & 3 & 1 \\ 3 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

6406533459061. ✖ None of these.

Question Number : 70 Question Id : 6406531021299 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Consider a function $f(x,y) = 3x^2 - 6xy + y^2 + 4$. Using gradient descent algorithm, with an initial guess of (1, 2) and learning rate of $\frac{1}{t^2}$, where $t = 1, 2, 3, \ldots$. What will be the value of (x, y) after two iterations?

Options :

6406533459066. ✖ $x = 7, y = 4$

6406533459067. ✔ $x = 2.5, y = 12.5$

6406533459068. ✖ $x = 4, y = 6$

6406533459069. ✖ $x = 3.5, y = 10$

Question Number : 71 Question Id : 6406531021300 Question Type : MCQ Correct Marks : 3

Question Label : Multiple Choice Question

Find the point on the line defined by $y = 2x + 1$ that is closest to the point $(4, 0)$.

Options :

6406533459070. ✖ (3,7)

6406533459071. ✔ (0.4, 1.8)

6406533459072. ✖ (0,0)

6406533459073. ✖ (-1,0)

Sub-Section Number :	4
Sub-Section Id :	640653155067
Question Shuffling Allowed :	No

Question Id : 6406531021292 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (72 to 74)

Question Label : Comprehension

Consider the matrix $A = \begin{pmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \end{pmatrix}$. Let the singular value decomposition (SVD) form of A be written as $A = Q_1 \sum Q_2^T$.

Based on the above data answer the given subquestions.

Sub questions

Question Number : 72 Question Id : 6406531021293 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
Which of the following options is/are true?
Options :

6406533459042. ✖ $Q_1 = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$

6406533459043. ✔ $Q_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$

6406533459044. ✖ $Q_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}$

6406533459045. ✖ $Q_1 = \sqrt{2} \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}$

Question Number : 73 Question Id : 6406531021294 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
Which of the following options is/are true ?
Options :

6406533459046. ✖ $Q_2^T = \frac{1}{\sqrt{6}} \begin{pmatrix} 2 & -1 & 1 \\ 0 & \sqrt{3} & \sqrt{3} \\ -\sqrt{2} & 1 & \sqrt{2} \end{pmatrix}$

6406533459047. ✖ $Q_2^T = \frac{1}{\sqrt{6}} \begin{pmatrix} 2 & -1 & 1 \\ 0 & \sqrt{3} & \sqrt{3} \\ -\sqrt{2} & -\sqrt{2} & -\sqrt{2} \end{pmatrix}$

6406533459048. ✖ $Q_2^T = \frac{1}{\sqrt{6}} \begin{pmatrix} 2 & -1 & 1 \\ 0 & \sqrt{3} & \sqrt{3} \\ \sqrt{2} & -\sqrt{2} & \sqrt{2} \end{pmatrix}$

6406533459049. ✔

$$Q_2^T = \frac{1}{\sqrt{6}} \begin{pmatrix} 2 & -1 & 1 \\ 0 & \sqrt{3} & \sqrt{3} \\ -\sqrt{2} & -\sqrt{2} & \sqrt{2} \end{pmatrix}$$

Question Number : 74 Question Id : 6406531021295 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
Which of the following options is true?
Options :

6406533459050. ✔ $\Sigma = \begin{pmatrix} \sqrt{3} & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$

6406533459051. ✖ $\Sigma = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$

6406533459052. ✖ $\Sigma = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$

6406533459053. ✖ $\Sigma = \begin{pmatrix} \sqrt{3} & 0 & 0 \\ 0 & \sqrt{3} & 0 \end{pmatrix}$

Sub-Section Number :	5
Sub-Section Id :	640653155068
Question Shuffling Allowed :	No

Question Id : 6406531021301 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (75 to 77)
Question Label : Comprehension

Consider the following dataset:

$$\left\{ \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 4 \\ 3 \end{pmatrix}, \begin{pmatrix} 6 \\ 5 \end{pmatrix} \right\}$$

Suppose we want to project the above dataset onto a 1 dimensional space using PCA.

Based on the above data answer the given subquestions.

Sub questions

Question Number : 75 Question Id : 6406531021302 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

The sample covariance matrix C for the given dataset is

Options :

6406533459074. ✔ $C = \frac{1}{3}\begin{pmatrix} 8 & 8 \\ 8 & 8 \end{pmatrix}$

6406533459075. ✖ $C = \frac{1}{3}\begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}$

6406533459076. ✖ $C = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

6406533459077. ✖ $C = \frac{1}{3}\begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}$

Question Number : 76 Question Id : 6406531021303 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

The eigenvalues of the covariance matrix are

Options :

6406533459078. ✖ $\lambda_1 = 16, \lambda_2 = 0$

6406533459079. ✖ $\lambda_1 = 4, \lambda_2 = 0$

6406533459080. ✖ $\lambda_1 = 1, \lambda_2 = 0$

6406533459081. ✔ $\lambda_1 = \frac{16}{3}, \lambda_2 = 0$

Question Number : 77 Question Id : 6406531021304 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

If $\tilde{x}_1, \tilde{x}_2$ and $\tilde{x}_3$ are the projections of the data points onto the first principal component, then which among the following is correct?

Options :

6406533459082. ✔ $\tilde{x}_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \tilde{x}_2 = \begin{pmatrix} 4 \\ 3 \end{pmatrix}, \tilde{x}_3 = \begin{pmatrix} 6 \\ 5 \end{pmatrix}$

6406533459083. ✖ $\tilde{x}_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \tilde{x}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \tilde{x}_3 = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$

6406533459084. ✖ $\tilde{x}_1 = \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix}, \tilde{x}_2 = \begin{pmatrix} 2/2 \\ 2/2 \end{pmatrix}, \tilde{x}_3 = \begin{pmatrix} 3/2 \\ 1/2 \end{pmatrix}$

6406533459085. ✖ $\tilde{x}_1 = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \tilde{x}_2 = \begin{pmatrix} 4 \\ 0 \end{pmatrix}, \tilde{x}_3 = \begin{pmatrix} 6 \\ 0 \end{pmatrix}$

Sub-Section Number :	6
Sub-Section Id :	640653155069
Question Shuffling Allowed :	Yes

Question Number : 78 Question Id : 6406531021305 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Which of the following statements is true about the given matrix M?

$$M = \begin{pmatrix} 2 & 4 \\ 1 & 2 \end{pmatrix}$$

Options :

6406533459086. ✖ M is an invertible matrix.

6406533459087. ✔ M is a singular matrix.

6406533459088. ✖ The determinant of M is 4.

6406533459089. ✔ M is diagonalizable.

Sub-Section Number :	7
Sub-Section Id :	640653155070
Question Shuffling Allowed :	Yes

Question Number : 79 Question Id : 6406531021306 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question

Which of the following statements is true about the matrix $A = \begin{pmatrix} 1 & 4 \\ 4 & 16 \end{pmatrix}$?

Options :

6406533459090. ✖ A is positive definite.

6406533459091. ✔ A is positive semi-definite.

6406533459092. ✖ A is negative definite.

6406533459093. ✖ A is negative semi-definite.

Java

Section Id :	64065373890
Section Number :	6
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	16
Number of Questions to be attempted :	16
Section Marks :	100
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653155071
Question Shuffling Allowed :	No

Question Number : 80 Question Id : 6406531021307 Question Type : MCQ
Correct Marks : 0
Question Label : Multiple Choice Question
THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : PROGRAMMING CONCEPTS USING JAVA (COMPUTER BASED EXAM)"